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Programme : Bachelor in Information Technology (Honours) (Internet of Things)
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TABLE OF CONTENTS

1. EXECUTIVE SUMMARY	1
2. INTRODUCTION	2
2.1. Background	2
2.2. Problem Statement	2
2.3. Objectives	3
2.4. Scope and Limitations	3
3. DATASET DESCRIPTIONS	4
3.1. Data Source	4
3.2. Dataset Overview	4
3.3. Variable Description	5
3.4. Data Complexity Justification	7
4. METHODOLOGY	8
4.1. Data Processing	8
4.1.1. Handling Missing Values	8
4.1.2. Duplicate Removal	10
4.1.3. Feature Engineering	11
4.1.4. Data Encoding and Scaling	12
4.2. Exploratory Data Analysis (EDA)	13
4.2.1. Univariate Analysis	13
4.2.2. Bivariate and Multivariate Analysis	14
4.3. Machine Learning Approach	15
4.4. Tools and Libraries	15
5. RESULTS AND FINDINGS	16
5.1. Exploratory Data Analysis (EDA) Results	16
5.1.1. Booking Cancellation Distribution	16
5.1.2. Hotel Type Distribution and Cancellation	16
5.1.3. Monthly Booking Trends	17

5.1.4.	Top Booking Countries.....	17
5.1.5.	Market Segment and Distribution Channel	18
5.1.6.	Deposit Type Analysis	18
5.1.7.	Customer and Guest Profile.....	18
5.2.	Correlation Analysis Results	19
5.3.	Machine Learning Model Results	20
5.3.1.	Model Accuracy Comparison	20
5.3.2.	Random Forest Feature Importance.....	20
5.3.3.	Key Findings Summary	21
6.	DISCUSSION	22
6.1.	Interpretation of Results	22
6.2.	Actionable Insights.....	23
6.3.	Model Performance Discussion.....	25
6.4.	Limitations	25
7.	CONCLUSION.....	26
8.	TEAM CONTRIBUTIONS	27
9.	REFERENCES	28
10.	APPENDICES	29
	Appendix A: Python Libraries and Environment	29
	Appendix B: Dataset Variables Full List.....	30
	Appendix C: Model Evaluation Details	32

1.EXECUTIVE SUMMARY

This report provides detailed analysis of data collected from hotel booking demand – using real-world data available on Kaggle.com. The data set comprises of 119,390 rows of hotel booking information across two types: City Hotels and Resort Hotels, and 32 columns including booking information, guest demographics, pricing, and cancellation status.

The initial aim of this project is to investigate what booking patterns are, what are the main reasons for bookings being cancelled, and create machine learning models to predict the likelihood of a booking being cancelled. There were three classification models developed and evaluated namely Logistic Regression, Decision Tree and Random Forest.

The major conclusions gained from this analysis are:

- The overall cancellation rate is 37.04%, while the City Hotels have a much higher cancellation rate (41.73%) than Resort Hotels (27.76%).
- The further ahead a booking is the greater chance of cancellation, with Lead Time being the most important predictor for this.
- Nearly all bookings that include a non-refundable deposit are cancelled (99.36%), thus showing an inverse correlation between non-refundable deposits and cancellation rate.
- August and July are the busiest booking months and most profitable revenue months, January is the quietest.
- Online Travel Agents (OTAs) have the biggest share in the booking channel, accounting for 47.29% of all bookings and the largest revenue share.
- Random Forest and Decision Tree showed the highest accuracy (81.77% and 81.73% respectively) compared to Logistic Regression (77.35%).

From these findings, a number of recommendations were put forward for implementation, such as having a tougher policy for paying guests with long term bookings, marketing more in low demand periods and offering better loyalty programmes for repeat bookings which account for just 3.2% of bookings.

2.INTRODUCTION

2.1. Background

The hotel and hospitality sector is undoubtedly one of the most data driven industries in the world. Hotels gather a lot of booking information daily such as pricing, guest profile, booking channels and cancelation trends. A combination of this data and analytical techniques can provide a multitude of competitive benefits, such as better utilization, improved revenue management, and minimizing the risk of last-minute cancellations.

Bookings cancellations are a significant challenge for hotels on both the operations and financial sides. If a guest decides to cancel, the hotel could end up with an empty room and lost income, particularly if the cancellation is too close to the guest's arrival. By identifying patterns and predictors of cancellation, hotels can implement proactive strategies like an overbooking policy, dynamic pricing, and changing their deposit requirements.

The Hotel Booking Demand dataset, one of the most popular datasets available in Kaggle, is used in this project, which consists of true booking records of two hotel categories in Portugal: City Hotel and Resort Hotel. This data set has been utilized in numerous academic and industry data science projects, making it appropriate for this analysis.

2.2. Problem Statement

Although data analytics has become more common in the hospitality industry, many hotels are still using gut instincts and not data-driven approaches to bookings and cancellations. This project solves the following problem:

What are the key determinants of hotel booking cancellations and is it possible to forecast if a booking will be cancelled or not with a machine learning model to help hotel take preventive action?

2.3. Objectives

The specific objectives of this data analysis project are:

- I. To explore and understand the structure, quality, and distribution of the hotel bookings dataset.
- II. To perform comprehensive Exploratory Data Analysis (EDA) to identify patterns, trends, and anomalies in booking behaviour.
- III. To engineer meaningful features and prepare the data for machine learning modelling.
- IV. To build and evaluate three classification models Logistic Regression, Decision Tree, and Random Forest for predicting booking cancellations.
- V. To derive actionable insights and business recommendations based on the analysis findings.

2.4. Scope and Limitations

The analysis is based on the booking data of two hotel properties (City Hotel and Resort Hotel) for the period 2015–2017. Analysis is only performed based on variables available in the data set and does not include a competitor price, macro-economic factors or real-time demand signals.

There are some caveats to note:

- I. There were 31,994 duplicate rows in the data that were removed, potentially reducing the representativeness of the sample;
- II. The company column had 94.3% of missing values and was zero-filled, which may introduce noise into the data; and (3) model predictions are generalised from historical patterns and may not apply to future booking behaviour under different market conditions.

3.DATASET DESCRIPTIONS

3.1. Data Source

The dataset used in this project was obtained from Kaggle.com, the world's largest data science community platform. Full details are as follows:

Field	Details
Dataset Name	Hotel Booking Demand
Kaggle URL	https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand
Original Research	Antonio, Almeida & Nunes (2019), Data in Brief, Elsevier
File Format	CSV (Comma-Separated Values)
License	CC0: Public Domain
Date Accessed	19 May 2026

3.2. Dataset Overview

After loading, the raw dataset contains the following characteristics:

Attribute	Value
Total Records (Rows)	119,390
Total Variables (Columns)	32
Duplicate Rows (removed)	31,994
Records after Deduplication	87,396
Target Variable	is_canceled (Binary: 0 = Not Cancelled, 1 = Cancelled)
Time Period Covered	2015 – 2017
Hotel Types	City Hotel, Resort Hotel

3.3. Variable Description

The dataset contains 32 variables spanning booking details, guest demographics, room and pricing information, and outcome labels. Key variables are described below:

```
[6]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 119390 entries, 0 to 119389
Data columns (total 32 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   hotel                                119390 non-null  str
1   is_canceled                          119390 non-null  int64
2   lead_time                           119390 non-null  int64
3   arrival_date_year                   119390 non-null  int64
4   arrival_date_month                  119390 non-null  str
5   arrival_date_week_number            119390 non-null  int64
6   arrival_date_day_of_month           119390 non-null  int64
7   stays_in_weekend_nights             119390 non-null  int64
8   stays_in_week_nights                119390 non-null  int64
9   adults                              119390 non-null  int64
10  children                            119386 non-null  float64
11  babies                              119390 non-null  int64
12  meal                                119390 non-null  str
13  country                             118902 non-null  str
14  market_segment                      119390 non-null  str
15  distribution_channel                 119390 non-null  str
16  is_repeated_guest                   119390 non-null  int64
17  previous_cancellations               119390 non-null  int64
18  previous_bookings_not_canceled       119390 non-null  int64
19  reserved_room_type                   119390 non-null  str
20  assigned_room_type                   119390 non-null  str
21  booking_changes                      119390 non-null  int64
22  deposit_type                         119390 non-null  str
23  agent                               103050 non-null  float64
24  company                             6797 non-null   float64
25  days_in_waiting_list                 119390 non-null  int64
26  customer_type                        119390 non-null  str
27  adr                                  119390 non-null  float64
28  required_car_parking_spaces          119390 non-null  int64
29  total_of_special_requests            119390 non-null  int64
30  reservation_status                   119390 non-null  str
31  reservation_status_date              119390 non-null  str
dtypes: float64(4), int64(16), str(12)
memory usage: 29.1 MB
```

Variable	Data Type	Description
hotel	Object	Hotel type: 'City Hotel' or 'Resort Hotel'
is_canceled	Integer (Binary)	Target variable: 1 = Cancelled, 0 = Not Cancelled
lead_time	Integer	Number of days between booking and arrival date
arrival_date_month	Object	Month of arrival (January to December)
stays_in_weekend_nights	Integer	Number of weekend nights booked
stays_in_week_nights	Integer	Number of weekday nights booked
adults / children / babies	Integer / Float	Number of guests by age group
meal	Object	Meal type booked (BB, HB, FB, SC, Undefined)
country	Object	Country of origin of the guest (ISO 3166 code)
market_segment	Object	Market segment (e.g., Online TA, Direct, Groups)
distribution_channel	Object	Booking distribution channel (TA/TO, Direct, etc.)
is_repeated_guest	Integer (Binary)	1 = Returning guest, 0 = New guest
previous_cancellations	Integer	Number of prior cancellations by the guest
adr	Float	Average Daily Rate — average revenue per occupied room
deposit_type	Object	Deposit type: No Deposit, Non Refund, Refundable
customer_type	Object	Type of booking: Transient, Contract, Group, Transient-Party
reservation_status	Object	Final reservation status (dropped before ML modelling)
total_guests*	Float (engineered)	Total number of guests = adults + children + babies
total_nights*	Integer (engineered)	Total stay length = weekend nights + week nights
total_cost*	Float (engineered)	Estimated total cost = ADR x total nights

3.4. Data Complexity Justification

This dataset satisfies the project's data complexity requirements on multiple dimensions:

- **Scale:** 119,390 records and 32 variables provide sufficient volume for statistical analysis and machine learning modelling.
- **Variable Diversity:** The dataset contains mixed data types including integer, float, categorical (object), and binary variables, requiring comprehensive preprocessing.
- **Missing Values:** Four variables contained missing data (children: 4, country: 488, agent: 16,340, company: 112,593), demanding different imputation strategies per variable.
- **Class Imbalance:** The target variable has a 63:37 split between non-cancelled and cancelled bookings, posing a classification challenge.
- **Feature Engineering Potential:** The dataset supports meaningful derived features (total_guests, total_nights, total_cost) that enhance model performance.
- **Domain Complexity:** Multiple business dimensions intersect pricing, seasonality, geography, customer behaviour, and distribution channels enabling rich multidimensional analysis.

4.METHODOLOGY

4.1. Data Processing

4.1.1.Handling Missing Values

Four variables contained missing values, each handled with an appropriate strategy:

```
[8]: missing_values = df.isnull().sum()

missing_values[missing_values > 0].sort_values(ascending=False)

[8]: company      112593
      agent       16340
      country      488
      children      4
      dtype: int64

[9]: missing_percentage = (df.isnull().sum() / len(df)) * 100

missing_percentage[missing_percentage > 0].sort_values(ascending=False)

[9]: company      94.306893
      agent      13.686238
      country      0.408744
      children      0.003350
      dtype: float64

[10]: df['children'] = df['children'].fillna(
        df['children'].median()
    )

    df['country'] = df['country'].fillna(
        df['country'].mode()[0]
    )

    df['agent'] = df['agent'].fillna(0)

    df['company'] = df['company'].fillna(0)

    print("Missing values handled successfully!")

Missing values handled successfully!
```

Variable	Missing Count	Missing %	Strategy Used	Justification
children	4	0.003%	Filled with median value	Very few missing; median robust to outliers
country	488	0.41%	Filled with mode (most frequent)	Categorical variable; PRT was dominant country
agent	16,340	13.69%	Filled with 0	No agent code implies direct/no agent booking
company	112,593	94.3%	Filled with 0	Near-total absence implies individual travellers

4.1.2. Duplicate Removal

The dataset contained 31,994 duplicate rows (approximately 26.8% of total records). These were identified using pandas `.duplicated()` and removed using `.drop_duplicates(inplace=True)`, resulting in a cleaner, deduplicated dataset for analysis.

```
[11]: df.isnull().sum().sum()

[11]: np.int64(0)

[12]: duplicates = df.duplicated().sum()

      print("Duplicate Rows:", duplicates)

Duplicate Rows: 32013

[13]: df.drop_duplicates(inplace=True)

      print("Duplicates removed successfully!")
      print("New Shape:", df.shape)

Duplicates removed successfully!
New Shape: (87377, 32)
```

4.1.3.Feature Engineering

Three new features were created to enhance analytical and predictive power:

Total Guests Feature

```
[15]: df['total_guests'] = (  
    df['adults'] +  
    df['children'] +  
    df['babies']  
    )  
df[['adults', 'children', 'babies', 'total_guests']].head()
```

```
[15]:
```

	adults	children	babies	total_guests
0	2	0.0	0	2.0
1	2	0.0	0	2.0
2	1	0.0	0	1.0
3	1	0.0	0	1.0
4	2	0.0	0	2.0

Total Stay Nights

```
[16]: df['total_nights'] = (  
    df['stays_in_weekend_nights'] +  
    df['stays_in_week_nights']  
    )  
df[['stays_in_weekend_nights',  
    'stays_in_week_nights',  
    'total_nights']].head()
```

```
[16]:
```

	stays_in_weekend_nights	stays_in_week_nights	total_nights
0	0	0	0
1	0	0	0
2	0	1	1
3	0	1	1
4	0	2	2

Total Cost Feature

```
[17]: df['total_cost'] = (  
    df['adr'] *  
    df['total_nights']  
    )  
df[['adr', 'total_nights', 'total_cost']].head()
```

```
[17]:
```

	adr	total_nights	total_cost
0	0.0	0	0.0
1	0.0	0	0.0
2	75.0	1	75.0
3	75.0	1	75.0
4	98.0	2	196.0

New Feature	Formula	Purpose
total_guests	adults + children + babies	Captures overall party size for guest profile analysis
total_nights	stays_in_weekend_nights + stays_in_week_nights	Represents total length of stay
total_cost	adr x total_nights	Estimates total revenue value of each booking

Additionally, the arrival_date_month variable was converted to an ordered categorical type to ensure correct chronological ordering in visualisations and analyses.

4.1.4.Data Encoding and Scaling

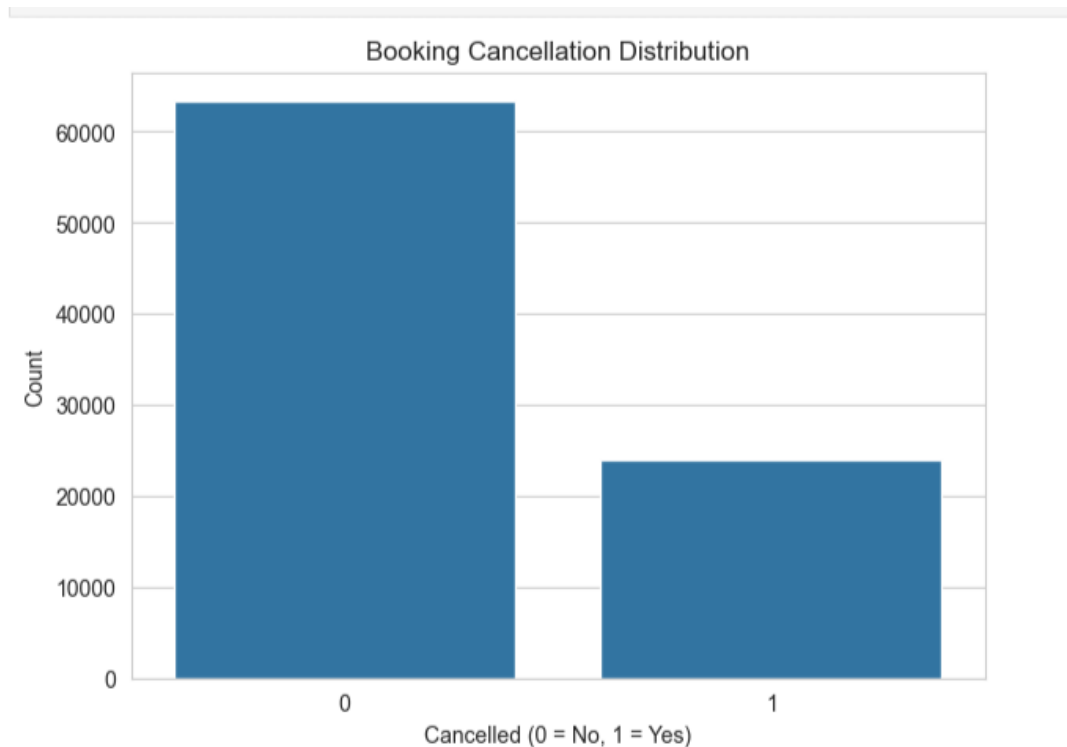
For machine learning preparation, the following transformations were applied:

- **Label Encoding:** All categorical and object-type variables were encoded to numerical values using scikit-learn's LabelEncoder. This covered variables such as hotel, country, meal, market_segment, distribution_channel, deposit_type, customer_type, and reserved_room_type.
- **Feature Scaling:** StandardScaler was applied to the training and test sets to normalise feature magnitudes, particularly important for Logistic Regression which is sensitive to feature scale.
- **Target Leakage Prevention:** The reservation_status and reservation_status_date columns were removed from the ML dataset prior to modelling, as these directly reveal the cancellation outcome and would cause data leakage.

4.2. Exploratory Data Analysis (EDA)

4.2.1. Univariate Analysis

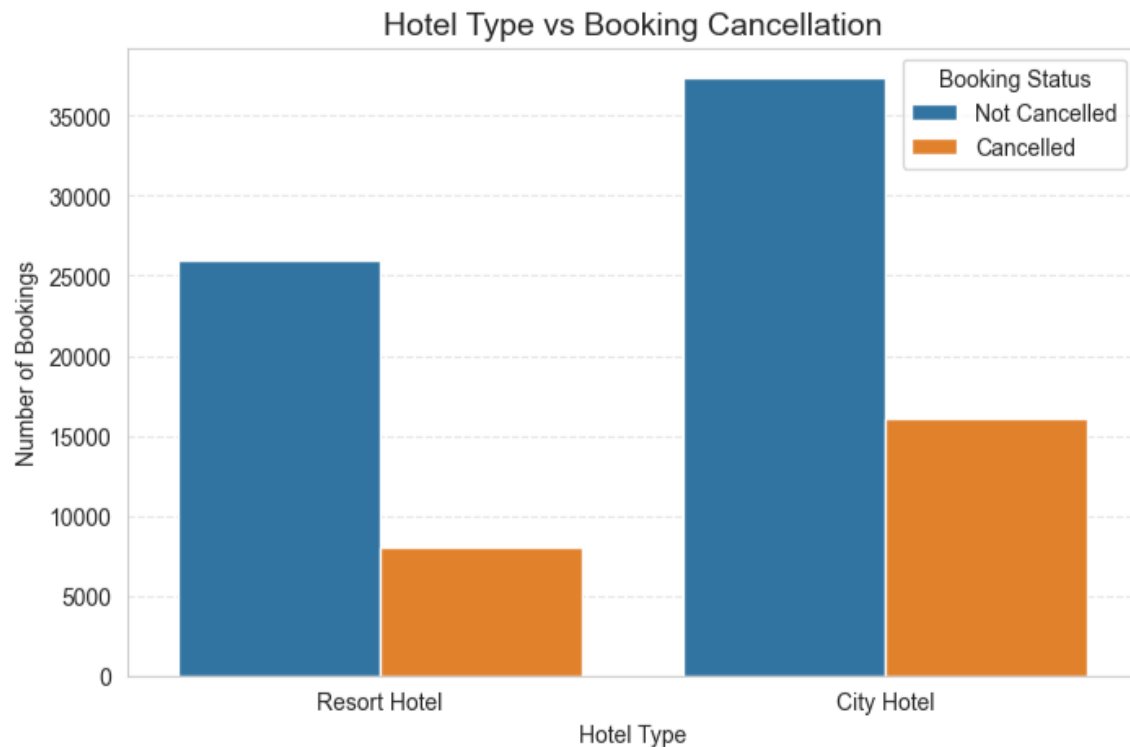
Individual variables were explored through count plots, histograms, pie charts, and bar charts to understand their distributions. Key observations:



- `is_canceled`: Imbalanced binary distribution 62.96% not cancelled vs 37.04% cancelled.
- `lead_time`: Right-skewed distribution with mean of 104 days, median of 69 days, and a maximum of 737 days. Most bookings are made within 160 days.
- `adr`: Distribution centred around the median of ~94.58, with notable outliers reaching up to 5,400. The 99th percentile filter was applied for boxplot visualisation.
- `total_nights`: Majority of stays are short median of 3 nights, with most stays between 2–4 nights.

4.2.2. Bivariate and Multivariate Analysis

Relationships between pairs of variables were examined to identify patterns relevant to the business questions:



- Hotel Type vs Cancellation: City Hotels exhibit a 41.73% cancellation rate versus 27.76% for Resort Hotels, suggesting structural differences in booking patterns.
- Deposit Type vs Cancellation: Non-Refund deposit bookings show a near-total 99.36% cancellation rate a counterintuitive finding that warrants investigation.
- Monthly Booking Trends: August (13,877 bookings) and July (12,661) are peak months; January (5,929) is the quietest month.
- Correlation Heatmap: lead_time ($r = 0.185$) and required_car_parking_spaces ($r = -0.184$) show the strongest linear correlations with cancellation.

4.3. Machine Learning Approach

Model	Algorithm Type	Key Parameters	Rationale
Logistic Regression	Linear, probabilistic	max_iter=500, random_state=42	Baseline model; fast, interpretable, works well with scaled features
Decision Tree	Non-linear, tree-based	max_depth=10, random_state=42	Captures non-linear relationships; depth capped to prevent overfitting
Random Forest	Ensemble, tree-based	n_estimators=100, max_depth=10, random_state=42	Reduces overfitting via bagging; provides feature importance rankings

The dataset was split 80/20 for training and testing with stratify=y to maintain class proportions. StandardScaler was applied for Logistic Regression; tree-based models were trained on unscaled data.

4.4. Tools and Libraries

Tool / Library	Version	Purpose
Python	3.x	Primary programming language
Pandas	Latest	Data loading, manipulation, and aggregation
NumPy	Latest	Numerical operations and array manipulation
Matplotlib	Latest	Base plotting and chart creation
Seaborn	Latest	Statistical visualisations (heatmaps, boxplots, countplots)
Scikit-learn	Latest	Machine learning models, preprocessing, and evaluation
Jupyter Notebook	Latest	Interactive development and code documentation environment
Kaggle.com	N/A	Dataset source and data hosting platform

5.RESULTS AND FINDINGS

5.1. Exploratory Data Analysis (EDA) Results

5.1.1.Booking Cancellation Distribution

Out of 119,390 total booking records, 44,224 bookings were cancelled (37.04%) and 75,166 were not cancelled (62.96%). This significant cancellation rate more than one in three bookings highlights the scale of the problem for hotel revenue management.

Status	Count	Percentage
Not Cancelled (0)	75,166	62.96%
Cancelled (1)	44,224	37.04%
Total	119,390	100%

5.1.2.Hotel Type Distribution and Cancellation

City Hotels account for a large majority of bookings at 66.4% (79,330 records) versus Resort Hotels at 33.6% (40,060 records). However, City Hotels also have a substantially higher cancellation rate:

Hotel Type	Total Bookings	Share	Cancellation Rate	Average ADR	Average Stay (Nights)
City Hotel	79,330	66.4%	41.73%	\$105.30	2.98 nights
Resort Hotel	40,060	33.6%	27.76%	\$94.95	4.32 nights

Resort Hotels attract guests with longer stays and lower cancellation rates, suggesting a more committed guest profile likely leisure travellers with fixed holiday plans. City Hotels cater to business and short-stay travellers who may be more flexible.

5.1.3.Monthly Booking Trends

Monthly booking volumes reveal a clear seasonal pattern:

Month	Bookings	Month	Bookings
January	5,929	July	12,661
February	8,068	August	13,877 (Peak)
March	9,794	September	10,508
April	11,089	October	11,160
May	11,791	November	6,794
June	10,939	December	6,780

The summer months of July and August represent peak demand, driven by leisure travel. January, November, and December represent the off peak period with the lowest booking volumes. Revenue analysis confirms August as the highest revenue month, generating an estimated total cost of \$7,944,717.

5.1.4.Top Booking Countries

Portugal (PRT) is the dominant source market with 48,590 bookings, reflecting the domestic nature of the dataset (the hotels are based in Portugal). The top 10 countries account for the majority of all bookings:

Rank	Country Code	Bookings
1	PRT (Portugal)	48,590
2	GBR (United Kingdom)	12,129
3	FRA (France)	10,415
4	ESP (Spain)	8,568
5	DEU (Germany)	7,287
6	ITA (Italy)	3,766
7	IRL (Ireland)	3,375
8	BEL (Belgium)	2,342
9	BRA (Brazil)	2,224
10	NLD (Netherlands)	2,104

5.1.5. Market Segment and Distribution Channel

Online Travel Agents (Online TA) dominate both booking volume and revenue:

Market Segment	Bookings	Share	Estimated Revenue
Online TA	56,477	47.3%	\$23,942,048
Offline TA/TO	24,219	20.3%	\$8,151,913
Groups	19,811	16.6%	\$4,669,637
Direct	12,606	10.6%	\$5,093,028
Corporate	5,295	4.4%	\$774,295
Others	982	0.8%	—

5.1.6. Deposit Type Analysis

The deposit type distribution reveals an important paradox:

Deposit Type	Bookings	Share	Cancellation Rate
No Deposit	104,641	87.6%	28.38%
Non Refund	14,587	12.2%	99.36%
Refundable	162	0.1%	22.22%

The near-total cancellation rate for Non-Refund deposits is counterintuitive. This is likely because Non-Refund deposits are often used for heavily discounted rates through OTA channels, where travellers book speculatively and cancel if their plans change despite losing the deposit. This finding suggests the Non-Refund policy may be ineffective as a cancellation deterrent.

5.1.7. Customer and Guest Profile

Transient customers dominate at 75.1% of all bookings. Most guests (87.8%) are non-family travellers (2 or fewer guests). Repeat guests represent only 3.2% of bookings (3,810 out of 119,390), indicating a significant opportunity to improve guest loyalty and retention.

5.2. Correlation Analysis Results

Pearson correlation coefficients between numerical variables and the target variable (is_canceled) reveal the following strongest relationships:

Variable	Correlation with is_canceled	Direction
lead_time	0.185	Positive — longer lead time, higher cancellation risk
total_cost	0.139	Positive — higher cost bookings slightly more likely to cancel
Adr	0.128	Positive — higher daily rate linked to higher cancellation
total_guests	0.100	Positive — larger groups slightly more likely to cancel
required_car_parking_spaces	-0.184	Negative — parking request signals committed guest
total_of_special_requests	-0.121	Negative — special requests indicate engaged guest
booking_changes	-0.094	Negative — modifications suggest intent to stay
is_repeated_guest	-0.090	Negative — returning guests much less likely to cancel

5.3. Machine Learning Model Results

5.3.1. Model Accuracy Comparison

Model	Accuracy	Precision (Cancel)	Recall (Cancel)	F1-Score (Cancel)
Logistic Regression	77.35%	66%	36%	0.47
Decision Tree	81.77%	70%	58%	0.64
Random Forest	81.73%	79%	46%	0.58

Decision Tree achieved the highest overall accuracy at 81.77%, slightly ahead of Random Forest (81.73%). Logistic Regression performed notably lower at 77.35%, primarily due to its limited ability to capture non-linear relationships in this dataset.

5.3.2. Random Forest Feature Importance

The Random Forest model identifies the 15 most influential features for predicting cancellation. The top 5 are:

Rank	Feature	Importance Score	Interpretation
1	lead_time	0.1347	Earlier booking = higher cancellation risk
2	market_segment	0.1188	Booking channel strongly influences cancellation behaviour
3	total_of_special_requests	0.1123	Guests with requests are more committed
4	country	0.1119	Guest origin correlates with cancellation patterns
5	required_car_parking_spaces	0.0803	Parking requests = higher guest commitment

5.3.3.Key Findings Summary

The following are the most significant findings from the complete analysis:

- I. 37.04% of all hotel bookings are cancelled, representing a major revenue risk for hotels.
- II. City Hotels suffer a 41.73% cancellation rate — nearly 14 percentage points higher than Resort Hotels at 27.76%.
- III. Lead time is the single most important predictor: bookings made far in advance are substantially more likely to be cancelled.
- IV. Non-Refund deposit bookings cancel at 99.36%, revealing that this deposit policy is not functioning as a cancellation deterrent.
- V. Only 3.2% of guests are repeat visitors — indicating very low guest loyalty and retention.
- VI. August generates the highest revenue (~\$7.9M estimated), while January is the weakest month.
- VII. Online TA is the dominant booking channel (47.3% of bookings, \$23.9M estimated revenue).
- VIII. Decision Tree and Random Forest models both achieve ~81.7% accuracy in predicting cancellations.

6.DISCUSSION

6.1. Interpretation of Results

This cancellation rate of 37% aligns with industry statistics indicating that hotel cancellation rates have dramatically increased as additional online booking platforms have come to market, providing free cancellation policies. As suggested, the higher cancellation rate at City Hotels (41.73%) may be attributed to the type of customers who stay in these establishments, as the City Hotels have a larger number of business travellers and short-break tourists whose plans tend to be more flexible, which implies that their switching costs are low.

The finding that the lead time positively correlates with cancellation ($r = 0.185$) is consistent with the principle of temporal discounting from behavioural economics; that guests who book in advance at a high lead time are more likely to cancel due to uncertainty about what will happen later. This will have direct impact on revenue management bookings with very long lead times should be marked as higher risk.

One of the most striking findings is that the percentage of deposit bookings that are cancelled is near total (99.36%) for Non-Refund deposits. Although it is not obvious, one would think that non-refundable deposits would make it more difficult to cancel a contract, this is probably due to the nature of this booking group. Non-refund rates are generally provided at reduced rates via OTA. These rates are tentative and can be cancelled if plans change without penalty. This indicates that the existing Non-Refund policy is not on track to fulfill its purpose.

This dominance is attributable to the fact that the hotels included in the data are mostly Portuguese (PRT) based hotels. But the UK, France, Spain and Germany are well represented, too, suggesting a strong international profile of guests, which has implications for providing multilingual service, marketing and distribution channels.

6.2. Actionable Insights

Based on the analysis findings, the following evidence-based recommendations are proposed for hotel management:

- I. Use variable deposit requirements based on lead time: If a booking is made over 90 days ahead then ask for a partial or full deposit to lessen the likelihood of cancellation. The data reveals that there is a strong relation between the higher cancellation probability and the longer lead times.
- II. Hotels need to consider whether they are attracting the right customer for the Non-Refund deposit strategy or whether it is just attracting speculative bookings, given the cancellation rate of 99.36% for Non-Refund bookings. Explore the option of implementing partial refund policies with Non-Refund, to find a balance between deterring cancellations and providing guest flexibility.
- III. Proactively intervene with the predictive model: Score incoming bookings based on the probability of cancellation using the Decision Tree or RF model. Increased retention communications (such as early check-in upgrades, targeted offers) can be sent based on bookings to enhance confirmation rates for high-risk bookings.
- IV. Make the most of peak months with revenue strategies: hotels should enforce no-cancellation policies nearer the arrival date during peak months, ensure dynamic pricing during those periods and ensure they focus on the guaranteed booking.

- V. Consider marketing outside the busy season (in January, November and December). These months could be especially good for occupancy if these marketing strategies are implemented: targeted promotions, loyalty rewards and package deals.
- VI. There's a lot of room for repeat guests – 3.2% of the ones that come back. There is significant untapped room for repeat guests with 3.2% repeat guests. Repeat guests do in fact have lower cancellation rates (is_repeated_guest correlation of -0.090 with cancellation) and so investment in loyalty is financially warranted.
- VII. Guest with special request are less likely to cancel with a negative correlation - 0.121: Engaged guests are less likely to cancel than non engaged guests. HOSLETS must actively engage guests to make their stay more personalized in order to establish commitment.

6.3. Model Performance Discussion

The result of the best balance of accuracy (81.77%) and recall for cancelled bookings (58%) was the Decision Tree model which is the recommended model for deployment. While Random Forest achieved similar accuracy (81.73%), its lower recall for cancellations (46%) means it misses more actual cancellations a more costly error from a revenue management perspective.

Logistic Regression was only 77.35 accurate with very low Recall for the Cancelled Booking (36%). This is an indication that there is considerable non-linear complexity in the cancellation prediction problem that cannot be captured with a linear model.

All three models need to be improved, especially in terms of recall for the cancelled class. Potential future research involves the use of oversampling techniques like SMOTE to deal with class imbalance, hyperparameter optimization using GridSearchCV, and more sophisticated ensemble classifiers like XGBoost or LightGBM.

6.4. Limitations

Several limitations should be considered when interpreting these results:

- The dataset covers only two hotel properties in Portugal, limiting the geographical and operational generalisability of findings.
- The time period (2015–2017) may not reflect current market dynamics, particularly post-pandemic shifts in booking behaviour.
- The company variable was retained with 94.3% zero-filled values, which may dilute its analytical value.
- External variables such as competitor pricing, weather, local events, and economic conditions are not captured in the dataset.
- The Label Encoding applied to high-cardinality categorical variables (e.g., country with 170+ values) may introduce unintended ordinal relationships in linear models.

7.CONCLUSION

This project accomplished a full data analysis of a hotel bookings demand from a real world hotel booking data set provided by Kaggle consisting of 119,390 booking reports. The project successfully tackled the primary research question on what influences the hotel cancellation rate and the prediction accuracy of hotel cancellations by performing systematic data preprocessing, exploratory data analysis and modelling using machine learning.

The overall cancellation rate was identified as 37.04% with City Hotels being much more impacted by cancellation rates than Resort Hotels. The most important factors for cancellation were found to be: lead time, market segment, number of special requests and country of origin. Deposit type analysis revealed an alarming discovery that the policy was ineffective as the bookings of Non-Refund deposits cancel at near total rate (99.36%).

Three machine learning models were created and tested. The accuracy of Decision Tree and Random Forest was around 81.7% which is a significant improvement over the baseline of 77.35% accuracy of Logistic Regression. The RF model further gave insights on the feature importance which was in coherence with the results obtained from an EDA which confirmed the interpretation of the key features.

Key recommendations are: redesign of the Non-Refund deposit policy, deploy the predictive model for proactive retention, increase marketing efforts in low demand periods, invest in a guest loyalty programme to increase the low repeat guest rate of 3.2%. These interventions can go a long way in effectively mitigating revenue loss due to cancellation and enhance overall operational efficiency.

Advanced ensemble model such as XGBoost, external market data for more contextual modelling, real-time cancellation risk scoring system to be integrated with the hotel's property management system are some possible future development.

8. TEAM CONTRIBUTIONS

Member Name	Student ID	Key Contributions	Contribution %
AHMAD ASLAM MUBARAK BIN ASWAN	52224224355	Data collection, dataset description, report writing	25%
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MUHAMMAD MUSTAQIM BIN MUKHTAR	52224224333	Discussion, conclusions, insights, video presentation coordination	25%

Video Presentation Link (Youtube)

[Video Presentation](#)

GitHub Repository Link:

[Project Materials](#)

9. REFERENCES

All references are formatted in APA 7th Edition.

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10. APPENDICES

Appendix A: Python Libraries and Environment

The following libraries were imported at the start of the Jupyter Notebook:

- pandas, numpy — data manipulation and numerical operations
- matplotlib.pyplot, seaborn — data visualisation
- sklearn.model_selection.train_test_split — data splitting
- sklearn.preprocessing.LabelEncoder, StandardScaler — encoding and scaling
- sklearn.linear_model.LogisticRegression — Logistic Regression model
- sklearn.tree.DecisionTreeClassifier — Decision Tree model
- sklearn.ensemble.RandomForestClassifier — Random Forest model
- sklearn.metrics: accuracy_score, confusion_matrix, classification_report — evaluation metrics

Appendix B: Dataset Variables Full List

#	Variable Name	Type	Description
1	hotel	Object	Hotel type (City Hotel / Resort Hotel)
2	is_canceled	Int64	Target: 1 = cancelled, 0 = not cancelled
3	lead_time	Int64	Days between booking and arrival
4	arrival_date_year	Int64	Year of arrival
5	arrival_date_month	Category	Month of arrival
6	arrival_date_week_number	Int64	Week number of arrival
7	arrival_date_day_of_month	Int64	Day of month of arrival
8	stays_in_weekend_nights	Int64	Weekend nights booked
9	stays_in_week_nights	Int64	Week nights booked
10	adults	Int64	Number of adults
11	children	Float64	Number of children
12	babies	Int64	Number of babies
13	meal	Object	Meal plan type
14	country	Object	Guest country of origin
15	market_segment	Object	Market segment
16	distribution_channel	Object	Distribution channel
17	is_repeated_guest	Int64	1 = returning guest
18	previous_cancellations	Int64	Prior cancellations by guest
19	previous_bookings_not_canceled	Int64	Prior confirmed bookings
20	reserved_room_type	Object	Room type code reserved
21	assigned_room_type	Object	Room type code assigned
22	booking_changes	Int64	Number of amendments made
23	deposit_type	Object	Deposit type
24	agent	Float64	Travel agent ID
25	company	Float64	Company/agency ID
26	days_in_waiting_list	Int64	Days on waiting list
27	customer_type	Object	Customer booking type

28	adr	Float64	Average Daily Rate
29	required_car_parking_spaces	Int64	Parking spaces requested
30	total_of_special_requests	Int64	Number of special requests
31	reservation_status	Object	Final status (removed before ML)
32	reservation_status_date	Object	Date of status (removed before ML)

Appendix C: Model Evaluation Details

Logistic Regression – Classification Report

Class	Precision	Recall	F1-Score	Support
0 (Not Cancelled)	0.79	0.93	0.86	12,671
1 (Cancelled)	0.66	0.36	0.47	4,805
Accuracy	—	—	0.77	17,476
Macro Avg	0.73	0.65	0.66	17,476
Weighted Avg	0.76	0.77	0.75	17,476

Decision Tree – Classification Report

Class	Precision	Recall	F1-Score	Support
0 (Not Cancelled)	0.85	0.91	0.88	12,671
1 (Cancelled)	0.70	0.58	0.64	4,805
Accuracy	—	—	0.82	17,476
Macro Avg	0.78	0.75	0.76	17,476
Weighted Avg	0.81	0.82	0.81	17,476

Random Forest – Classification Report

Class	Precision	Recall	F1-Score	Support
0 (Not Cancelled)	0.82	0.95	0.88	12,671
1 (Cancelled)	0.79	0.46	0.58	4,805
Accuracy	—	—	0.82	17,476
Macro Avg	0.80	0.71	0.73	17,476
Weighted Avg	0.81	0.82	0.80	17,476